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HPE Alletra Storage MP X10000 Data Intelligence User Guide

Part Number: 20-ALMPX10K-DI-UG-ED1

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HPE Alletra Storage MP X10000 Data Intelligence User Guide

Abstract

This guide describes HPE X10000 Data Intelligence and demonstrates how to create pipelines and collections, process object events, query metadata, enable RAG workflows, and automate enrichment and search through APIs and UI tools to operationalize intelligent data processing. This guide is intended for customers after the server node and software have been installed and configured in your environment.

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Revision history

Part number	Publication date	Edition	Summary of changes
20-ALMPX10K-DI-UG-ED1	February 2026	1	First release of HPE Alletra Storage MP X10000 Data Intelligence.

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HPE X10000 Data Intelligence

The demand for more intelligent, integrated, and efficient methods to store, manage, and extract value from data is reshaping the storage market landscape. HPE X10000 Data Intelligence is designed to meet these new customer challenges.

The HPE X10000 Data Intelligence solution (hardware and software) is an optional component for the HPE Alletra Storage MP X10000 storage system. It is designed to create AI-ready datasets by using advanced AI and analytics technologies to extract, enrich, and store metadata and embeddings from X10000 objects and files in real time. The extracted information is stored in a vector DB or in Apache Iceberg tables for further searches.

Data intelligence transforms the X10000 into a storage infrastructure that embeds data processing and enrichment directly in the storage layer, so unstructured objects can be parsed, annotated with custom metadata, indexed, and queried without first copying data to an external analytics platform.

Subtopics

[HPE X10000 Data Intelligence features](#)

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[Glossary and terminology](#)

HPE X10000 Data Intelligence features

- Zero-code pipeline creation: Create RAG/embedding and metadata-extraction pipelines by configuration (Swagger/UI/JSON). No container coding or custom orchestration is required.
- Event-driven, online, storage-integrated processing: as an object is created, modified, or deleted in the X10000 bucket for Data Intelligence, object events are published (Kafka) and processed automatically by HPE X10000 Data Intelligence pipelines.
- Creation of two pipeline families:
 - RAG/embedding pipelines: chunking, embedding generation, vector indexing and similarity search for retrieval-augmented generation.
 - Metadata-extraction pipelines: extract system and user S3/object metadata and write results to open table formats.
- Synchronized lifecycle and security: Derived intelligence (embeddings, metadata) inherits the same authentication/authorization and is synchronized with object lifecycle (creations and deletions happen in real time).
- Open, queryable storage of results: Embeddings stored in a vector store for semantic search. Structured outputs are written to open table formats (Iceberg) so results are searchable via SQL, dashboards, and external tools.
- APIs and SDK for integration: Standardized APIs and a public SDK (GitHub) let applications access embeddings, tables, and search results and supports custom application integration.
- Minimal data duplication and efficient design: Designed to avoid unnecessary object duplication into vector DBs and keep derived intelligence aligned with original objects.
- Managed, containerized runtime: The HPE X10000 Data Intelligence software automatically instantiates and manages containerized services (model runtimes, orchestration, API gateway, collection manager) so you do not manage low-level components.
- Model and collection management: An expandable model library lets you select an AI model to use in pipelines from a provided library of models. In a future release, the ability to bring your own models for custom enrichment will be added.
- Rich search capabilities: Similarity (vector) search, SQL/semantic search over metadata, RAG chat/dashboard application included.

HPE X10000 Data Intelligence use cases

- Retrieval-augmented generation (RAG)/Knowledge base: Ingest manuals, documentation, and knowledge articles, generate embeddings and enable semantic search and chat to help technicians or support staff quickly find solutions.
- Legal, compliance and eDiscovery: Index and semantically search large sets of regulatory and legal documents across jurisdictions to surface relevant compliance information.
- Metadata-driven analytics and dashboarding: Extract S3/system/user metadata into open tables (Iceberg) to build SQL queries and dashboards (examples: bucket sizes, file-type distributions, ingestion trends, geolocation maps).
- Image and scientific data cataloging and search: Extract and index metadata (for example, geocoordinates, tags, clinical labels) to find and group images (for example, georeferenced imagery and Alzheimer image sets) for research.
- Quality inspection and root-cause analysis: Capture and index metadata or extracted attributes (for example, valve defect level) to monitor production quality and query historical evidence.
- Fast similarity search across unstructured assets: Use embeddings and vector similarity to find similar documents, images, or data fragments for deduplication, discovery, or content-based retrieval.
- Operational automation and event-driven enrichment: Automatically run pipelines on object create/delete events to keep metadata and embeddings synchronized with object lifecycle for real-time workflows.
- Secure, policy-aware search and access: Provide searches and RAG results that respect the same authentication/authorization and lifecycle rules as the underlying objects.
- Hybrid application integration: Expose intelligence via APIs/SDK so custom applications, chat UIs, or MCP agents can use metadata, vectors, and RAG capabilities programmatically.

Glossary and terminology

Term	Definition
Bucket	An S3 bucket of an HPE Alletra Storage MP X10000 storage system.
Collection	Logical domain for identifying a set of object metadata that is extracted and queried. There can be one or more S3 user buckets which form a collection. Each collection consists of the following: • One or more buckets associated with the collection • One pipeline
Data Intelligence API Gateway	Management gateway for orchestrating Data Intelligence workflows on the HPE Alletra Storage MP X10000 storage system. It provides users with endpoints to create collection objects, pipeline and run search queries for Vector data and Metadata.
Embedding Model	Machine Learning (ML) model used to extract embeddings from the object data. A Model may be associated with a Pipeline Instance. This association is created when the pipeline is instantiated and cannot be modified for its lifetime. The embeddings generated from a model persist within the collection corresponding to the pipeline.
Event Endpoint	An endpoint to which events related to objects within an HPE Alletra Storage MP X10000 cluster are published. Events can be of different types and represent state changes in the lifetime of an object (for example, creation, overwrite, or delete).
Event Filter	Logical grouping of events published to an event endpoint. This can be expressed as a regular expression with wildcards (for example, <code>"*.pdf"</code>).
Apache Iceberg	Apache Iceberg is an open table format used in modern storage on S3, such as the HPE Alletra Storage MP X10000, that makes large-scale analytic data behave like a traditional database table but with the scalability of object storage.
LLM	Large Language Models

Term	Definition
Metadata	Some insight about the user data that is extracted and stored in an Iceberg table that enables the user to perform a structured query on the collection. Metadata can be both system-generated as well as user embedded.
Pipeline	Immutable representation of the processing required to extract metadata for Data Intelligence. A pipeline can be either the RAG or metadata type. The pipeline is an HPE X10000 Data Intelligence process, created by a user, to extract intelligent data from object data. A pipeline is composed of: • A model that identifies the embedding model to use for the pipeline or custom function.  NOTE Custom function is not currently supported but is expected to be supported in a future release. • One schema describing the logical structure of the metadata to be extracted.
Pipeline Orchestrator	Orchestrates the setup required to extract intelligent data from object data based on the pipeline definition added by user.
RAG	Retrieval-Augmented Generation (RAG) combines document retrieval with language generation to produce more accurate and context-aware responses. It enhances large language models by grounding their outputs in external, up-to-date information.
Schema	Defines the structure of metadata to be extracted or queried. This is equivalent to the schema that identifies the columns/fields within a database table.
Swagger UI	Swagger UI is a web-based interface that allows user to interact with and visualize RESTful APIs defined using the OpenAPI Specification.
Trino	Trino is a distributed SQL query engine designed to query large data sets. It is used to perform read, write, modify, and delete operations on collections that manage metadata.

Getting started

Subtopics

[HPE X10000 Data Intelligence key components](#)

[Swagger UI navigation](#)

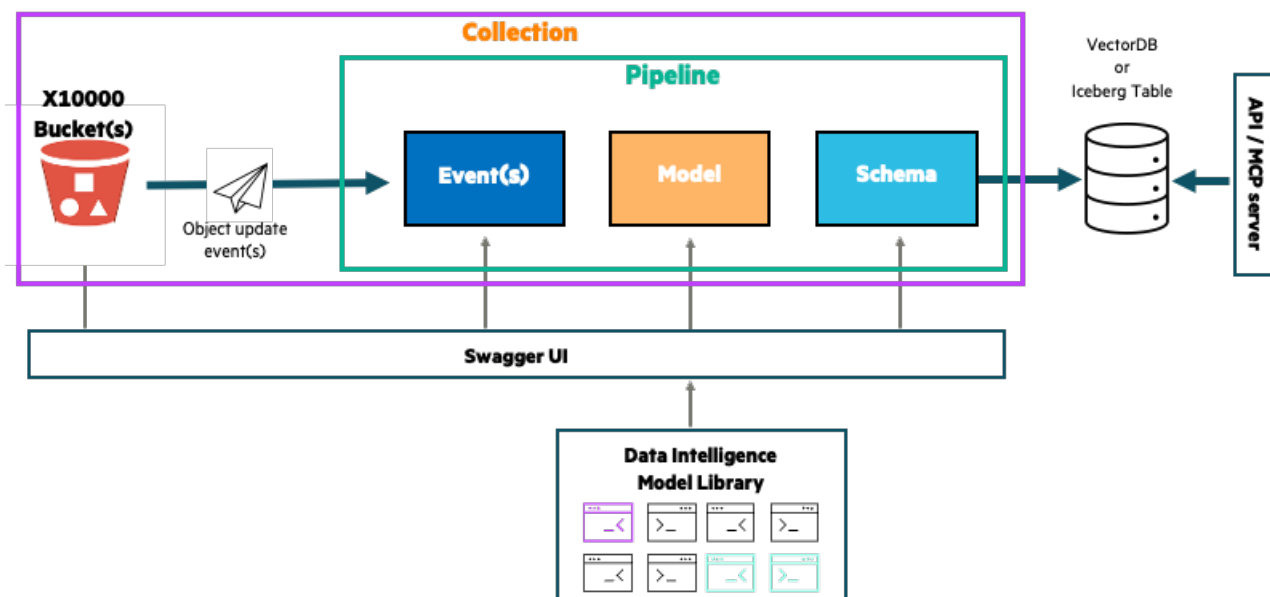
HPE X10000 Data Intelligence key components

The X10000 data intelligence software is built around the following key components:

- Pipeline
- Collection
- Buckets
- Models

- Schema
- Event
- Database

These components are related as shown in the following picture:



Subtopics

- [Pipeline](#)
- [Collection](#)
- [Models](#)
- [Schema](#)
- [Buckets](#)
- [Event](#)
- [Databases](#)

Pipeline

An HPE X10000 Data Intelligence pipeline is a configured, event-driven process that takes X10000 storage objects as input, runs analysis or enrichment models on them, and writes the derived intelligence to queryable stores so applications can find and use it.

The core elements of the HPE X10000 Data Intelligence pipeline are:

- **Event:** Objects create/delete events are produced when a bucket is marked for Data Intelligence. Events are delivered reliably via Kafka.
- **Model(s):** The pipeline selects one or more models (for example, embedding model, image/text extractor, LLM components) to analyze the object.
- **Schema:** Defines how the pipeline's outputs (metadata fields, vectors, and links to the source object) are stored.

The HPE X10000 Data Intelligence pipeline works as follows:

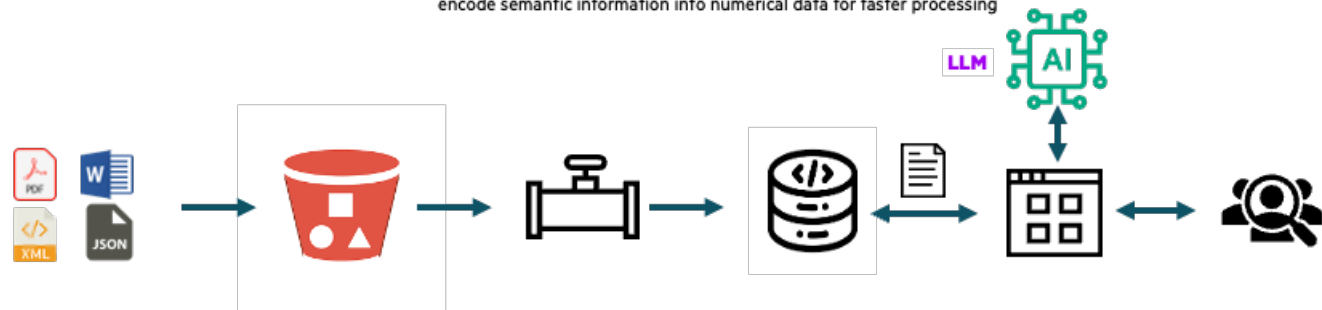
1. When an object event arrives, an event router maps it to the associated pipeline instance.
2. The pipeline performs tasks such as chunking text, running embeddings or extractors, and packaging results according to the schema.
3. Outputs are written into appropriate stores:
 - embeddings into a vector store for similarity search
 - structured metadata into open table formats (Iceberg) for SQL/dashboarding

Currently, HPE X10000 Data Intelligence supports creation of two types of pipelines:

- **Retrieval-augmented generation (RAG):** This pipeline enables you to create RAG pipelines and "chat" with the data stored in the bucket using simple configuration. It also streamlines RAG deployments by providing data synchronization and consistent managed access control, while minimizing vector database data duplication and enhancing security through aligned data and embedded security levels.
- **Metadata Extraction and Search:** This pipeline enables the creation or extraction of metadata from unstructured data and stores it in an open table format. This companion data can be accessed using native SQL query engines like Spark, Trino, or text-to-SQL to generate insights from it.

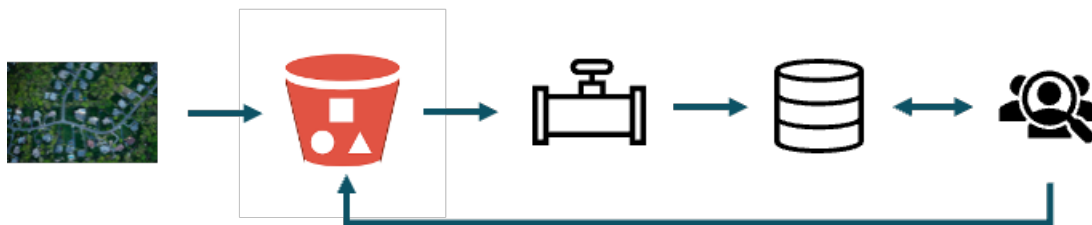
Retrieval-augmented generation (RAG)

encode semantic information into numerical data for faster processing



Metadata Extraction & Search

Capture, index, and handle metadata to accelerate access time to the object



Collection

A collection is the logical binding that connects a pipeline to one or more storage buckets. It represents the unit you query and manage in HPE X10000 Data Intelligence.

Key aspects:

- A collection is a logical combination of a pipeline and one or more buckets (collection = pipeline + bucket(s)). You build a pipeline (which specifies the triggering events, the model(s) to execute, and the output schema) and then link one or more buckets to that pipeline to create a collection.
- Event routing: When an object event (create/delete) is generated for a bucket, the event router dispatches it to the pipeline associated with the collection, so the configured processing runs automatically.
- Query target: Searches and queries are performed against a collection. Results come from the pipeline's output schema (embeddings in the vector store or structured fields in Iceberg/open tables).
- Reusability and scaling: A single pipeline can be reused by adding additional buckets to its collection, letting you expand processing to more datasets without redefining pipeline logic.
- Lifecycle and security alignment: Outputs produced for a collection (metadata, embeddings) are synchronized with the source objects and inherit the storage system's access controls, enabling creations and deletions to stay consistent.

Models

In HPE X10000 Data Intelligence, a model (sometimes referred to as an embedded model) is a selectable software component (often an AI or machine-learning element) that a pipeline uses to analyze or enrich objects as events occur. Models are the processing modules that perform the actual transformation of object data into insights.

Key aspects

- Pipeline selection: When you create a pipeline, you pick one or more models from the collection (model) library. The pipeline routes incoming object events to the selected model(s) for processing.
- Library and extensibility: HPE X10000 Data Intelligence maintains a collection manager/model library where built-in models live. You can add custom models so pipelines can run customer-specific inference.
- Runtime and containerization: Models are executed inside managed, containerized runtimes (the platform instantiates the required containers and manages them), hiding runtime complexity from users.
- Task mapping: Models perform chunking, embedding generation, metadata extraction, or LLM inference as defined by the pipeline configuration and schema.

Default models

The HPE X10000 Data Intelligence node includes two default models:

- `nomic-performance-model`
 - model name: nomic-embed-text
 - capabilities: embedding
 - dimension: 768
 - maximumTokens: 8192
 - version: v1.5
- `bge-versatile-model`
 - model name: bge-m3
 - capabilities: embedding
 - dimension: 1024
 - maximumTokens: 8192
 - version: 567m



IMPORTANT

Creating and deleting models is not currently supported.

Schema

In HPE X10000 Data Intelligence, a schema specifies the structure and format for storing intelligence generated by a pipeline. Meaning, how the extracted metadata and embedding vectors from the source object are organized and stored in the HPE X10000 Data Intelligence databases for querying and use.

The schema defines what fields the pipeline will write (such as system metadata, user metadata, object tags, embedding vectors, links to the source object, chunk IDs, timestamps, etc.), and how those fields are represented.

Default schemas

The HPE X10000 Data Intelligence node includes two default schemas:

- `default-metadata-schema`
- `default-rag-schema`



IMPORTANT

Only the default schemas are supported. Creating and deleting schemas is not currently supported.

Buckets

In the context of HPE X10000 Data Intelligence, a bucket is the storage container on the HPE Alletra Storage MP X10000 storage system that holds the objects or files you want the platform to analyze and enrich.

A bucket is configured (marked) as a HPE X10000 Data Intelligence bucket to enable processing for its objects (event notifier enable).

Event

An event in HPE X10000 Data Intelligence is a storage object lifecycle message (for example, object create or delete) produced by an X10000 bucket that has been marked for Data Intelligence. Events are the triggers that cause the platform to analyze and enrich objects.

How event works

- **Source and types:** Events are emitted when objects are inserted or removed from a configured Data Intelligence bucket (creates and deletes).
- **Delivery and reliability:** Events are delivered to the HPE X10000 Data Intelligence service via a Kafka queue, so they are reliably captured and not lost.
- **Routing and handling:** The event router/MoC component inspects each event and identifies the pipeline instance associated with the source bucket (collection = pipeline + bucket). The pipeline then invokes the configured model(s) to run tasks such as chunking, metadata extraction, embedding generation, or LLM steps.
- **Outcomes and synchronization:** The processing results are written back to the platform's stores (a vector DB for embeddings, Iceberg/open tables for structured metadata). Because events drive both create and delete actions, derived intelligence (embeddings and metadata) stays synchronized with the underlying objects and inherits the same lifecycle and access controls.

Events make the system fully automated and real-time. Moving a file into a bucket immediately triggers processing and makes enriched/searchable outputs available via APIs/SDK.

Databases

HPE X10000 Data Intelligence uses two primary database types to store the intelligence that pipelines produce:

- **Vector database (vector store):** Stores embedding vectors produced by embedding models, so similarity searches (semantic/RAG) can be performed. It is used at query time to find the nearest vectors or matching chunks for retrieval-augmented generation or similarity search.
- **Structured metadata database (open table/Iceberg tables):** Stores extracted system/user/application metadata and other structured fields, making the metadata queryable with SQL and usable for dashboards and analytics. Metadata is persisted in open table formats (Iceberg) so it is accessible by standard SQL tools and external systems.

Both databases live alongside the HPE Alletra Storage MP X10000 storage system (outputs are written back into the X10000 environment) and are accessible via the platform’s API/SDK.

Lifecycle and security are synchronized with the source objects. Creations and deletions trigger pipeline actions, and corresponding entries in the vector DB and metadata tables are kept synchronized and respect the same access controls.

Swagger UI navigation

Main view

When first opened, the Swagger UI displays the endpoints (REST API) by group (for example, collections, pipelines).

authn	authentication and authorization	>
collections	collection of buckets with associated model pipeline	>
pipelines	pipelines for data processing	>
models	models for data processing	>
schemas	schemas for data processing	>
search		>



NOTE

The search endpoints are meant to programmatically integrate external applications with the HPE X10000 Data Intelligence. They are intended for developers only.

API endpoints

Click on a group to expand it, for example, the collections group. Each endpoint shows its HTTP method, path, description, and required parameters.

collections collection of buckets with associated model pipeline 

GET

/api/v1/collections

Get Collections



POST

/api/v1/collections

Create a new collection



GET

/api/v1/collections/{name}

Get collection by name



DELETE

/api/v1/collections/{name}

Deletes a collection



POST

/api/v1/collections/{name}/assignBuckets

Assign list of buckets to a collection



POST

/api/v1/collections/{name}/unassignBuckets

Unassign list of buckets from a collection



Each endpoint represents a specific operation you can perform, such as managing datasets and metadata, triggering analytics workflows, and retrieving insights. Within the Swagger UI, these endpoints are organized by functional groups, making it easy to explore how different parts of the system work together.

Each endpoint includes:

- A description of what the operation does
- Input parameters and expected payloads
- Response formats, including success and error cases
- Interactive execution with the Try it out feature

Request and response details

When you click the button to the left of the endpoint name (for example, POST or GET), the endpoint panel is displayed.

The panel is divided into two parts:

- **Parameter:** You see the REST API input parameters (query, path, header, body). To input the parameter details, click Try it out. The input area become editable and you can modify the Request body. Click Execute to run the API.

POST /api/v1/collections Create a new collection

Create a new collection

Parameters

Cancel

No parameters

Request body

application/json

```
{  "name": "rag-collection-1",  "buckets": [    "bucket-1",    "bucket-2"  ],  "pipeline": "rag-pipeline"}
```

Execute

- **Response:** After clicking Execute, the system displays the response and the related return codes for the call:
 - The full request (URL, headers, body)
 - The actual server response (status code, body, and headers)

Responses		
Code	Description	Links
200	Successfully created the collection Media type application/json Controls Accept header. Example Value Schema <pre>{ "name": "rag-collection-1", "buckets": ["bucket-1", "bucket-2"], "pipeline": "rag-pipeline" }</pre>	No links
206	Partial success (some buckets invalid or not created) Media type application/json Example Value Schema <pre>{ "Status": "string", "invalidBuckets": ["string"], "validBuckets": ["string"], "collection": { "name": "rag-collection-1", "buckets": ["bucket-1", "bucket-2"], "pipeline": "rag-pipeline" } }</pre>	No links
401	Unauthorized	No links
405	Invalid input	No links
default	Internal Error Media type application/json Example Value Schema <pre>{ "code": "string", "details": "string" }</pre>	No links

HPE X10000 Data Intelligence configuration

HPE X10000 Data Intelligence software is fully API-driven, meaning all activities can be performed via API calls. Use the Swagger UI, which is a simple, no-code configuration user interface, to create and manage all HPE X10000 Data Intelligence aspects, such as pipeline and collection creation.

The Swagger UI is a visual, web-based interface that exposes a service's REST APIs in a human-readable format, enabling you to explore, test, and invoke API requests without writing code. With HPE X10000 Data Intelligence, you can execute real-time requests to manage, analyze, and monitor data intelligence workflows.

Subtopics

[Accessing the Swagger UI](#)

[Creating an S3 bucket using DSCC](#)

[Creating an S3 bucket using an S3 command](#)

[Creating a pipeline](#)

[Creating a collection](#)

Accessing the Swagger UI

Prerequisites

- Login requires the NodePort service (31443).
- If you are tunneling into the HPE X10000 Data Intelligence node, define port forwarding in your tunnel:

```
ssh -L 8051:localhost:31443 user@remote-host
```

- To verify the NodePort port, enter the following command on the HPE X10000 Data Intelligence node:

```
kubectl get svc -n hpe-di
```

About this task

To use the Swagger UI, use the login endpoint to complete authentication and authorization.

Procedure

1. Enter the following URL in the browser: `https://<nodeIP>:31443/api/doc#`

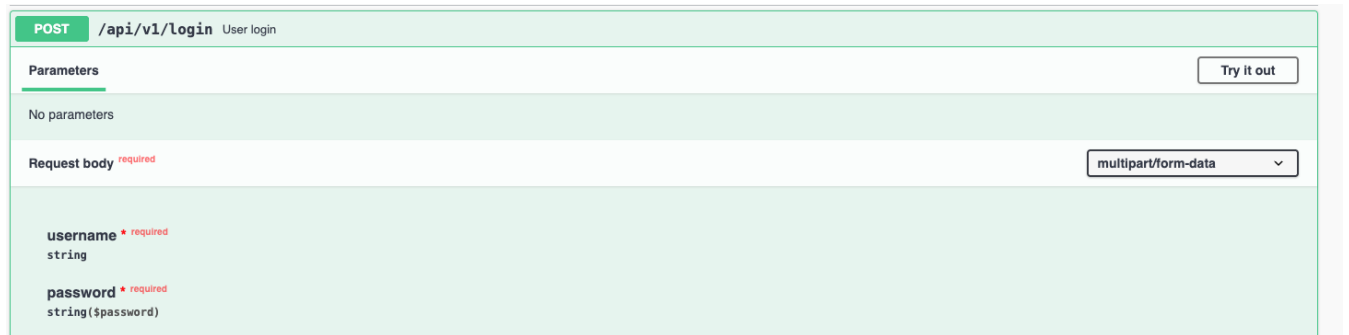
`<nodeIP>` is the IP address of the HPE X10000 Data Intelligence node.

2. Complete authentication:

- a. Select the `/api/v1/login` endpoint.



- b. Click POST to display the endpoint panel.



- c. Click Try it out and enter `admin` for both the username and password (this is the default). The username and password credentials are created when the HPE X10000 Data Intelligence was set up during installation.



NOTE

The username and password were set as `api-gateway.appCredentials.username` and `api-gateway.appCredentials.password` in the `values.yaml` file.

- d. Click Execute.

The response area is populated. If login is successful, an authorization token is displayed in the response body.

Server response

Code Details

200 Response body

```
{
```

```

    "Status": "Login successful",
    "Authorization": "Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJhZG1pbisInV4cC1MTc2OTlw
MzUwMCIsImJpbmJ1aXNoY
WQ1Q0hOTE0N3gVdG5xS5x-

NgY_gp4XqVP7pV5VFxrSOYeRpsqS7H1dHwpmZtwkrnG_PZBWGrvnp30U7KHJghF1g
5SjuX6wZQD679ZC8zWZr5dSvAJ0gLjS4HXSXD6nUf71azzn4kdp09QnGz3tQYDRNSAiB
jHGRBsRiF_yvsMhj1L1N7cN5UEqXsIAoKcc52

nY_gP4xgWTxyxeefbdBSNRFzkaJUzVPGf7LmFAuQEbeyvmdLvZWHUwSMrB8eH6J7g7T
Xuwx9ZPndYjFg9fiVFkE2IAWfAgc0xIYZcG7Teg"
  }

  Response headers
  authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJhZG1pbisInV4cC1MTc2OTlw
MzUwMCIsImJpbmJ1aXNoY
WQ1Q0hOTE0N3gVdG5xS5x-

NgY_gp4XqVP7pV5VFxrSOYeRpsqS7H1dHwpmZtwkrnG_PZBWGrvnp30U7KHJghF1g
5SjuX6wZQD679ZC8zWZr5dSvAJ0gLjS4HXSXD6nUf71azzn4kdp09QnGz3tQYDRNSAiB
jHGRBsRiF_yvsMhj1L1N7cN5UEqXsIAoKcc52

nY_gP4xgWTxyxeefbdBSNRFzkaJUzVPGf7LmFAuQEbeyvmdLvZWHUwSMrB8eH6J7g7T
Xuwx9ZPndYjFg9fiVFkE2IAWfAgc0xIYZcG7Teg
  content-length: 508
  content-type: application/json; charset=utf-8
  date: Thu, 22 Jan 2026 17:46:48 GMT
  server: HPE-X10000/api_gateway/v1
  x-process-time: 0.0526303986571172

```

e. Copy the Bearer token. It is required for authorization.

3. Complete authorization:

- a. Click the Authorize button, located at the top right of the Swagger UI.
- b. In the form that displays, paste the Bearer token in the Value field.



- c. Click Authorize.

You are now authenticated and authorized to operate Data Intelligence.

Available authorizations

×

bearerAuth (http, Bearer)

Authorized

Value: *****

Logout

Close

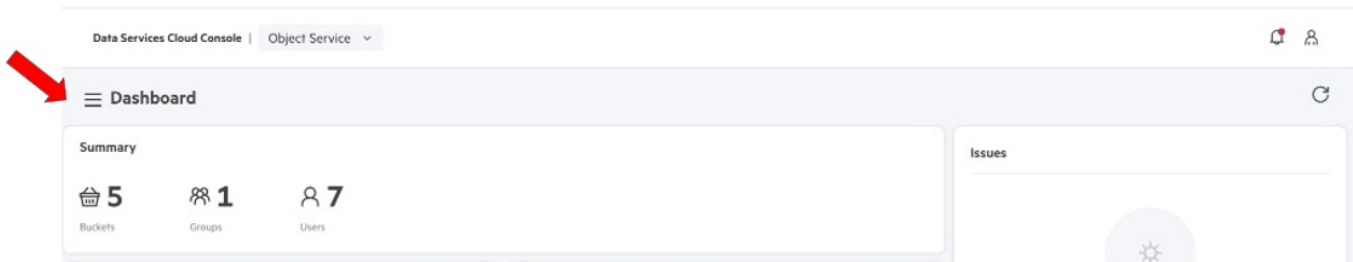
Creating an S3 bucket using DSCC

Prerequisites

The administrator must have access to manage users, access policy, and buckets in Object Service on the Data Services Cloud Console. For details on provisioning the privilege, see the Using Object Service articles collection at <https://www.hpe.com/support/GLObjectStorageArticles>.

Procedure

- 1. Access the Data Services Cloud Console at <https://common.cloud.hpe.com> and navigate to the X10000 dashboard.
- 2. Click the menu icon (next to Dashboard) and select Buckets.



The Buckets page is displayed.

Data Services Cloud Console | Object Service

🔔

👤

☰ Buckets

🔄

+

✎

✕

🔍

🏷️

📄

	Name	System	Storage Tier	Compression	Capacity Used (GiB)	Metadata	Performance (Last hour summary)			...
							Service Time	S3 Request	Data Transferred	
☐	tk-test	ftc-tme-s5359	HPE Alletra Storage MP	Disabled	1.07	1	0	0	0	

- 3. Click the + icon.
The Create Bucket wizard is displayed.
- 4. On the General tab, enter a name for the bucket (observe name limitations) and click Next.
- 5. On the Systems tab, select an HPE Alletra Storage MP X10000 system for this bucket to be provisioned. If there are multiple systems, the Object Service intuitively presents the best system for provisioning the bucket. Click Next.



Create Bucket



System

2 system(s) found. The recommended system is based on available free capacity.

ftc-tme-s5359

Model	HPE Alletra Storage MP X10000
Serial #	4UW0005359
Capacity	30.15 TiB free of 30.16 TiB

Recommended

ftc-tme-s4796

Model	HPE Alletra Storage MP X10000
Serial #	4UW0004796
Capacity	0.00 MiB free of 0.00 MiB

Cancel

Back

Next

- You do not have to enter or select anything on the Options tab. If preferred, enter key-value pairs in the Metadata field for administrative tracking. Click Next.



NOTE

It is not necessary to select S3 notification, as collection creation automatically activates this step.

Create Bucket



Progress bar: General (checked), System (checked), Options (active), Summary (unchecked)

☐ Enable Versioning ⓘ

☐ Enable Locking ⓘ

☒ Enable S3 Notification ⓘ

* Target

Select a Target ▼

* Topic ⓘ

Topic

Buttons: Cancel, Back, Next

7. On the Summary tab, review selections and click Create to provision the bucket.

Creating an S3 bucket using an S3 command

Prerequisites

You must be an authorized S3 user and have run `aws configure`.

Procedure

Enter the following command: `aws mb s3://<bucket_name> --endpoint-url=<endpoint_server_name>`

Creating a pipeline

About this task



NOTE

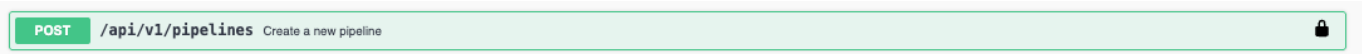
The HPE X10000 Data Intelligence node comes with two default, preconfigured pipelines:

- default-rag-pipeline
- default-metadata-pipeline

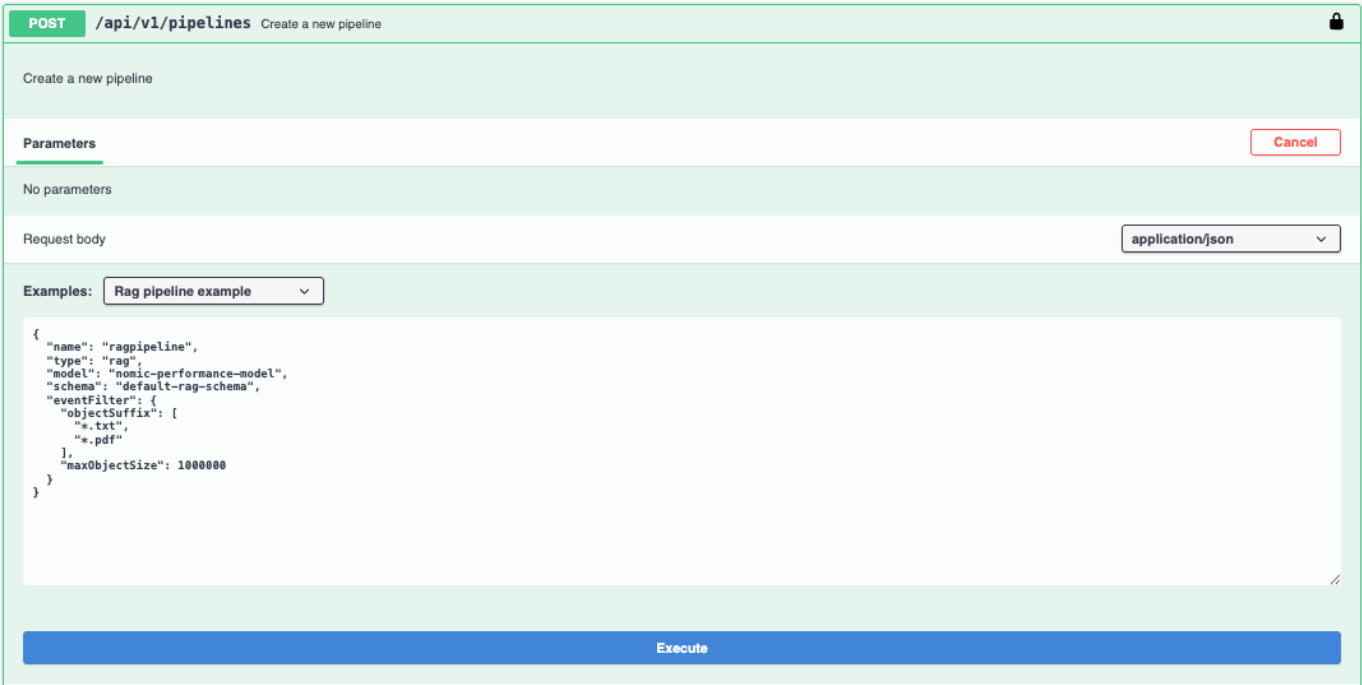
If you choose to use either of these defaults, you can skip this task.

Procedure

- 1. Open the pipelines endpoint group in the Swagger UI.
- 2. Select the Create a new pipeline endpoint.



- 3. Click POST to display the endpoint panel (Parameters and Response areas). By default, the Rag pipeline example is selected and the JSON file displayed in the following figure lists the parameters defined for this example.



- 4. Click Try it out to enable editing on the panel. Select a pipeline from the Examples dropdown menu and modify the values to create a custom pipeline.

The available values for the pipeline JSON file are:

JSON Input field	Description	Possible values
name	Name of the pipeline	Any string
type	Pipeline type	rag or metadata
model	The model used by the pipeline to analyze the objects	- nomic-performance-model - bge-versatile-model
schema	The model output schema stored in the database	- default-rag-schema - default-metadata-schema
eventFilter	List of filters	Supported object formats for rag are: *.pdf *.doc *.html *.json *.txt *.docx For metadata, you can select any object format (for example, *).
maxObjectsize	This parameter applies to the Rag pipeline only.	10 MB

- Click Execute to run the API.

If execution is successful, the response with `code=200` and a JSON file describing the new pipeline are displayed in the Response area.

Creating a collection

About this task

Creating a collection means assigning one or more buckets to a pipeline.



NOTE

Bucket-to-Collection is a many-to-many relationship. Meaning, a bucket can be associated with more than one collection at any time.

Procedure

- Open the collections endpoint group in the Swagger UI.
- Select the Create a new collection endpoint.

POST /api/v1/collections Create a new collection



3. Click POST to display the endpoint panel (Parameters and Response areas).

POST /api/v1/collections Create a new collection

Create a new collection

Parameters

No parameters

Request body

application/json

```
{
  "name": "rag-collection-1",
  "buckets": [
    "bucket-1",
    "bucket-2"
  ],
  "pipeline": "rag-pipeline"
}
```

Execute

The JSON file in the Request body area contains:

- The name of the collection you want to create ("name" :)
 - The pipeline you want to use ("pipeline" :)
 - The bucket(s) you want to associate to the pipeline ("Buckets" : [...])
4. Click Try it out to enable editing on the panel. Enter the applicable details.
 5. Click Execute to run the API.

If execution is successful, the Response area includes `code=200` and a JSON file describing the new pipeline. The details are similar to the following output:

```
{
  "name": "rag-collection-1",
  "buckets": [
    "bucket 1",
    "bucket 2"
  ],
  "pipeline": "rag-pipeline"
}
```

It is possible execution may be partially successful. Meaning, not all buckets were created. The message displayed might look like this:

```
{
  "Status": "string"
  "invalidBuckets": [
    "string"
  ],
  "validBuckets": [
    "string"
  ],
  "collection": [
    "name": "rag-collection-1",
    "buckets": [
```

```
"bucket-1",  
"bucket-2"  
],  
"pipeline": "rag-pipeline"  
}  
}
```

HPE X10000 Data Intelligence management with the Swagger UI

Using the Swagger UI provides a clear, hands-on view of how the system behaves—whether you are triggering data ingestion, managing metadata, launching analytics tasks, or reviewing processing results. Each operation is presented with detailed descriptions, sample payloads, and immediate responses, making it easy to validate functionality and understand the impact of each action.

Subtopics

- [Listing all pipelines](#)
- [Listing pipeline details by pipeline name](#)
- [Deleting a pipeline](#)
- [Listing all models](#)
- [Listing model details by model name](#)
- [Listing schemas](#)
- [Listing schema details by schema name](#)
- [Assigning buckets to an existing collection](#)
- [Unassigning buckets from a collection](#)
- [Listing all collections](#)
- [Listing collection details by collection name](#)
- [Deleting a collection](#)

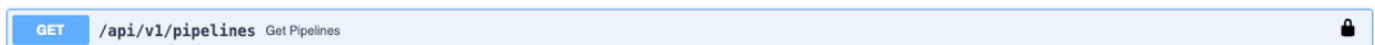
Listing all pipelines

About this task

Use the Get Pipelines endpoint to review all existing pipelines, understand their current status, and identify the workflows configured in the system. Accessing this list is often the first step before performing operations such as monitoring executions, triggering runs, or updating pipeline definitions.

Procedure

1. Open the pipelines endpoint group in the Swagger UI.
2. Select the Get Pipelines endpoint.



3. Click GET to display the endpoint panel (Parameters and Response areas).
4. Click Try it out to enable editing on the panel.

GET /api/v1/pipelines Get Pipelines

Get list of pipelines

Parameters

No parameters

Execute Clear

- Click Execute. The list of all configured pipelines is displayed in the Response area.

```
Request URL
https://172.21.161.65:31442/api/v1/pipelines

Server response
Code    Details
200     Response body
[
  {
    "id": "7d77c707-336f-4845-a968-8623becbfe98",
    "name": "default-rag-pipeline"
  },
  {
    "id": "6dd2d529-3602-4b1c-b37f-861ca5522157",
    "name": "default-metadata-pipeline"
  }
]
```

The HPE X10000 Data Intelligence node comes with two default pipelines:

- default-rag-pipeline
- default-metadata-pipeline

Listing pipeline details by pipeline name

About this task

Use the Get pipeline by name endpoint to view a pipeline's configuration, structure, and metadata without browsing the entire list. This endpoint is especially useful when you need to verify a pipeline's definition, troubleshoot an issue, or prepare for operations such as updates or executions.

Procedure

- Open the pipelines endpoint group in the Swagger UI.
- Select the Get pipeline by name endpoint.

GET /api/v1/pipelines/{name} Get pipeline by name

Try it out

- Click GET to display the endpoint panel (Parameters and Response areas).
- Click Try it out to enable editing on the panel.

GET
/api/v1/pipelines/{name}
Get pipeline by name

Returns a single pipeline

Parameters

Name
Description

name
required
string
(path)
Name of pipeline to return

Execute

Clear

- Enter the pipeline name and click **Execute**. The details of the pipeline are displayed in the Response area.

Server response

Code Details

200 Response body

```
{
  "name": "default-rag-pipeline",
  "type": "rag",
  "model": "nomic-performance-model",
  "eventFilter": {
    "maxObjects": 1000000,
    "objectSuffix": [
      ".pdf",
      ".ppt",
      ".docx",
      ".csv",
      ".html",
      ".json"
    ]
  },
  "schema": "default-rag-schema",
  "systemGenerated": true,
  "id": "151e9678-9b23-1b28-4777-5fce2497c2e9",
  "creationTime": "2025-12-07T21:11:46.079623Z",
  "lastModified": "2025-12-07T21:11:50.062882Z"
}
```

Response headers

content-length: 483

content-type: application/json; charset=utf-8

date: Mon, 02 Feb 2026 22:17:29 GMT

server: HPE-X10000/api_gateway/v1

x-process-time: 0.0295093390652468

Deleting a pipeline

About this task

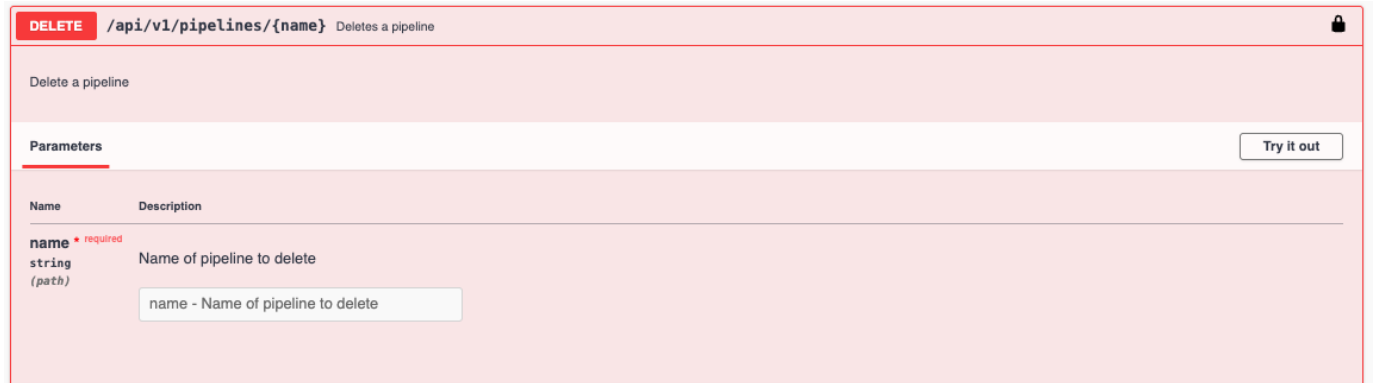
Deleting a pipeline allows you to permanently remove a pipeline that is no longer needed or has been replaced by a newer version.

Procedure

1. Open the pipelines endpoint group in the Swagger UI.
2. Select the Deletes a pipeline endpoint.



3. Click DELETE to display the endpoint panel (Parameters and Response areas).
4. Click Try it out to enable editing on the panel.

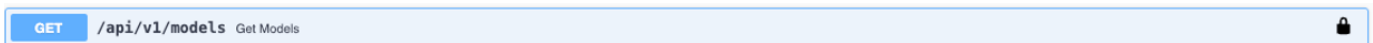


5. Enter the name of the pipeline and click Execute.
The result of the operation is displayed in the Response area.

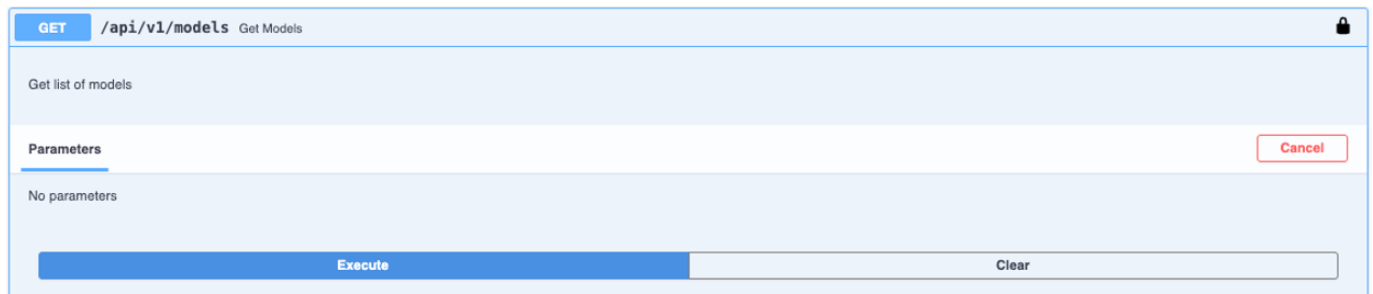
Listing all models

Procedure

1. Open the models endpoint group in the Swagger UI.
2. Select the Get Models endpoint.



3. Click GET to display the endpoint panel (Parameters and Response areas).
4. Click Try it out to enable editing on the panel.



5. Click Execute. The list of all configured models is displayed in the Response area.



Server response

Code Details

200 Response body

```
{
  "models": [
    {
      "id": "9c12ccc6-89a5-40f4-a3d6-61fcb5c1452",
      "name": "nomic-performance-model"
    },
    {
      "id": "c2362620-5cdf-4d39-b9fe-60fee8092a13",
      "name": "bge-versatile-model"
    }
  ]
}
```

Response headers

content-length: 174

content-type: application/json; charset=utf-8

date: Mon, 02 Feb 2026 22:26:48 GMT

server: HPE-X10000/api_gateway/v1

x-process-time: 0.06687939702533185

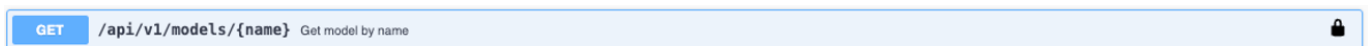
Listing model details by model name

About this task

Use the Get model by name endpoint to access a model's configuration, metadata, and current status without scanning the full list of models.

Procedure

1. Open the models endpoint group in the Swagger UI.
2. Select the Get model by name endpoint.



3. Click GET to display the endpoint panel (Parameters and Response areas).
4. Click Try it out to enable editing on the panel.



5. Enter the model name and click **Execute**. The details of the model are displayed in the Response area.

Server response

Code Details

200 Response body

```
{
  "name": "nomic-performance-model",
  "modelName": "nomic-embed-text",
  "capabilities": [
    "embedding"
  ],
  "dimension": 768,
  "maximumTokens": 8192,
  "version": "v1.5"
}
```

Response headers

content-length: 159

content-type: application/json; charset=utf-8

date: Mon, 02 Feb 2026 22:38:12 GMT

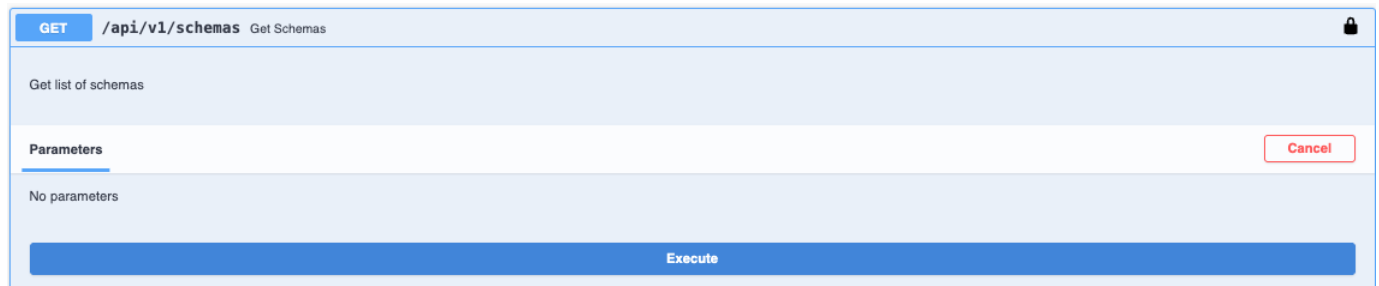
server: HPE-X10000/api_gateway/v1

x-process-time: 4.024538357959732

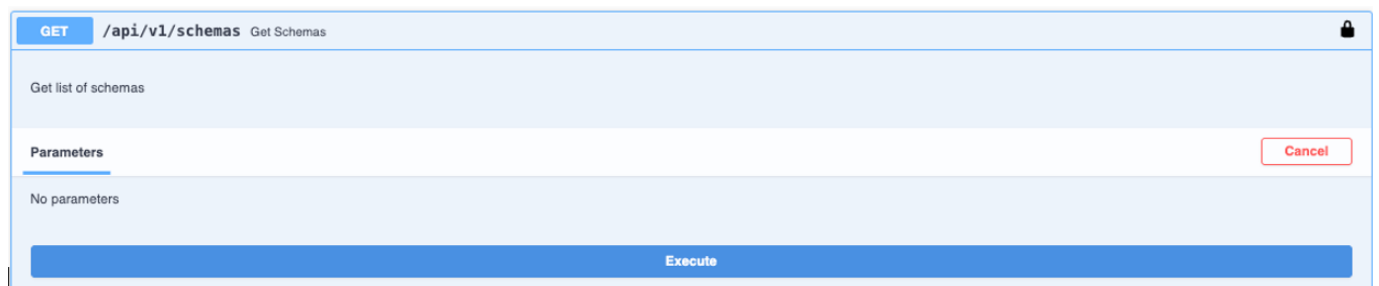
Listing schemas

Procedure

1. Open the schemas endpoint group in the Swagger UI.
2. Select the Get Schemas endpoint.



3. Click GET to display the endpoint panel (Parameters and Response areas).
4. Click Try it out to enable editing on the panel.



5. Click **Execute**. The list of all configured schemas is displayed in the Response area.

Server response
Code Details
200 Response body

```

{
  "schemas": [
    {
      "id": "37dbbeaf-2122-4477-81a6-a8cdb4573c65",
      "name": "default-metadata-schema"
    },
    {
      "id": "4ae3485c-9779-4d49-a8f7-59cf96d9436",
      "name": "default-rag-schema"
    }
  ]
}

```

Response headers
content-length: 174
content-type: application/json; charset=utf-8
date: Mon, 02 Feb 2026 22:48:10 GMT
server: HPE-X10000/api_gateway/v1
x-process-time: 2.0682411899324507

Listing schema details by schema name

Procedure

- 1. Open the schemas endpoint group in the Swagger UI.
- 2. Select the Get schema by name endpoint.
- 3. Click GET to display the endpoint panel (Parameters and Response areas).
- 4. Click Try it out to enable editing on the panel.

GET /api/v1/schemas/{name} Get schema by name

Returns a single schema

Parameters

Name	Description
name required string (path)	Name of schema to return

name - Name of schema to return

Execute

- 5. Enter the schema name and click Execute. The details of the schema are displayed in the Response area.

Server response
Code Details
200 Response body

```

{
  "name": "default-metadata-schema",
  "type": "metadata",
  "schema": [
    {
      "name": "object_key",
      "type": "varchar"
    },
    {
      "name": "bucket",
      "type": "varchar"
    },
    {
      "name": "version_id",
      "type": "varchar"
    },
    {
      "name": "size",
      "type": "bigint"
    },
    {
      "name": "last_modified",
      "type": "timestamp"
    },
    {
      "name": "content_type",
      "type": "varchar"
    }
  ]
}

```

Assigning buckets to an existing collection

Procedure

1. Open the collections endpoint group in the Swagger UI.
2. Select the Assign list of buckets to a collection endpoint.

POST /api/v1/collections/{name}/assignBuckets Assign list of buckets to a collection



3. Click POST to display the endpoint panel (Parameters and Response areas).

POST

/api/v1/collections/{name}/assignBuckets

Assign list of buckets to a collection

Assign list of buckets to a collection

Parameters

Cancel

Name	Description
name required string (path)	Name of collection to assign buckets rag-collection-1

Request body

application/json

```
{
  "buckets": [
    "bucket-1",
    "bucket-2"
  ]
}
```

Execute

- Enter the collection name and buckets to be assigned to the collection.
- Click Execute.

The results are displayed in the Response area. As described in [Creating a collection](#), there may be partial success if not all buckets are assigned to the collection.

Unassigning buckets from a collection

About this task

A bucket or list of buckets can be unassigned from an existing collection by specifying the collection name and the bucket names.

Procedure

- Open the collections endpoint group in the Swagger UI.
- Select the Unassign a list of buckets from a collection endpoint.

POST

/api/v1/collections/{name}/unassignBuckets

Unassign list of buckets from a collection

- Click POST to display the endpoint panel (Parameters and Response areas).

POST
/api/v1/collections/{name}/unassignBuckets
Unassign list of buckets from a collection

Unassign list of buckets from a collection

Parameters

Name	Description
name * required string (path)	Name of collection to unassign bucket

name - Name of collection to unassign bucke

Request body

application/json

```

{
  "buckets": [
    "bucket-1",
    "bucket-2"
  ]
}

```

Execute

- Enter the collection name and the buckets to unassign from it.
- Click Execute. The results of the operation are displayed in the Response area. As described in [Creating a collection](#), there may be partial success if not all buckets are unassigned from the collection.

Listing all collections

About this task

Use the Get collections endpoint to review all existing collections and identify the workflows configured in the system.

Procedure

- Open the collections endpoint group in the Swagger UI.
- Select the Get collections endpoint.
- Click GET to display the endpoint panel (Parameters and Response areas).
- Click Try it out to enable editing on the panel.

GET
/api/v1/collections
Get Collections

Get list of collections

Parameters

No parameters

Execute

5. Click Execute. The list of all configured collections is displayed in the Response area.

Server response

Code Details

200 Response body

```
[
  {
    "id": "0e7946c7-6aa4-41d3-8d77-a636f73f228c",
    "name": "ask-x10000-rag-collection-1"
  }
]
```

Response headers

content-length: 87

content-type: application/json; charset=utf-8

date: Mon, 02 Feb 2026 23:20:14 GMT

server: HPE-X10000/api_gateway/v1

x-process-time: 2.1388060110621154

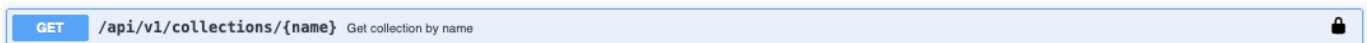
Listing collection details by collection name

About this task

You can retrieve detailed information about a specific collection by providing the name in the Swagger UI. Through this endpoint, you can quickly view a collection's configuration, structure, and metadata without browsing the entire list. It is especially useful when you need to verify a collection's definition or troubleshoot an issue.

Procedure

1. Open the collections endpoint group in the Swagger UI.
2. Select the Get collection by name endpoint.



3. Click GET to display the endpoint panel.
4. Click Try it out to enable editing on the panel.
5. Enter the name of the collection in the Parameters area.

The Swagger UI endpoint panel for the `GET /api/v1/collections/{name}` endpoint. It includes a description "Returns a single collection", a "Parameters" section with a "name" parameter (required, string, path) with the value "ask-x10000-rag-collection-1", and an "Execute" button at the bottom.

GET /api/v1/collections/{name} Get collection by name

Returns a single collection

Parameters

Name	Description
name * required string (path)	Name of collection to return

ask-x10000-rag-collection-1

Execute

6. Click Execute. Details similar to the following are displayed in the Response area.

Server response

Code

Details

200 Response body

```
{
  "name": "ask-x10000-rag-collection-1",
  "buckets": [
    "af-rag-demo-bucket"
  ],
  "pipeline": "default-rag-pipeline"
}
```

Response headers

content-length: 110

content-type: application/json; charset=utf-8

date: Mon, 02 Feb 2026 23:24:27 GMT

server: HPE-X10000/api_gateway/v1

x-process-time: 2.0227692099288106

Deleting a collection

About this task

Deleting a collection allows you to permanently remove a collection that is no longer needed or has been replaced by a newer version.

Procedure

- 1. Open the collections endpoint group in the Swagger UI.
- 2. Select the Deletes a collection endpoint.

DELETE

/api/v1/collections/{name}

Deletes a collection

🔒

3. Click DELETE to display the endpoint panel (Parameters and Response areas).

DELETE

/api/v1/collections/{name}

Deletes a collection

🔒

Delete a collection

Parameters

Cancel

Name	Description
name * required	Name of collection to delete
string	
(path)	
name - Name of collection to delete	

Execute

4. Enter the name of the collection and click Execute.

The result of the operation is displayed in the Response area.



Access methods for HPE X10000 Data Intelligence data

The data generated by HPE X10000 Data Intelligence is embeddings (vector representations) and structured metadata. The methods to access the data are:

- **RAG Chatbot:** HPE X10000 Data Intelligence provides a RAG (Retrieval Augmented Generation) chatbot to demonstrate typical embedding retrieval and data chat.
- **SQL-based access:** The extracted system and user metadata of HPE X10000 Data Intelligence are stored in open table formats, enabling you to run SQL queries, build dashboards, and utilize traditional analytics tools with the enriched dataset.

Subtopics

[Using the RAG chatbot](#)

[Using Trino CLI \(SQL-based access\)](#)

Using the RAG chatbot

Prerequisites

- Data Intelligence node is deployed.
- S3 credentials are known and available.
- The bucket being used should be populated with at least one file that matches the standards established during pipeline creation.

About this task

The Data Intelligence Retrieval Augmented Generation (RAG) chatbot is an AI assistant designed to provide accurate, context-aware answers by combining large language models with your organization's data. The RAG chatbot is accessed through the NodePort service.

Procedure

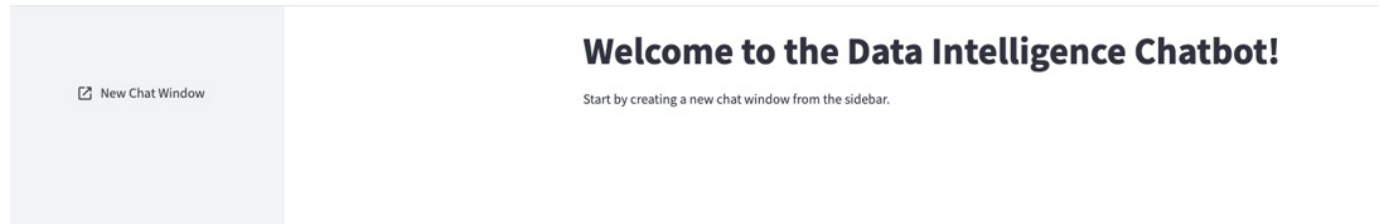
1. Open a browser and enter the URL: `http://<Server Name>:30501/`



NOTE

- The chatbot service is `http`, not `https`.
- The `<Server Name>` is the name of the Data Intelligence node.

The Welcome screen is displayed.



2. Complete the following steps to log in.
 - a. Click New Chat Window.

Enter S3 Credentials



Access Key

Secret Key



Submit

The Enter S3 Credentials screen is displayed.

3. Enter your S3 credentials and click Submit.



After credentials are verified, the chat icon is displayed in the top-right area of the screen.

4. Click the chat icon.

The chat configuration screen is displayed.

Enter config for the chat session



Select LLM

Llama 3.2 (Default)



Select Collection

Select a collection



Top K



System Prompt

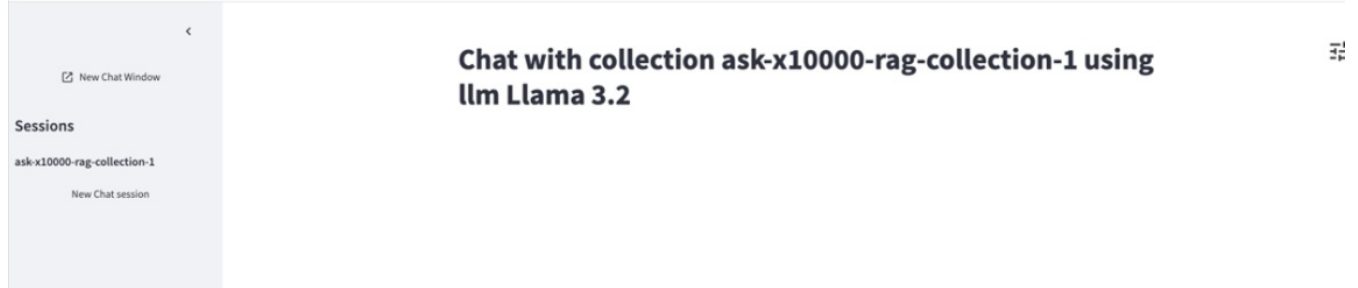
Enter the system prompt that will be applicable to the chatbot in this session. By default, no system prompt will be applied.

For example: 'Keep the answers short and concise.'

Submit

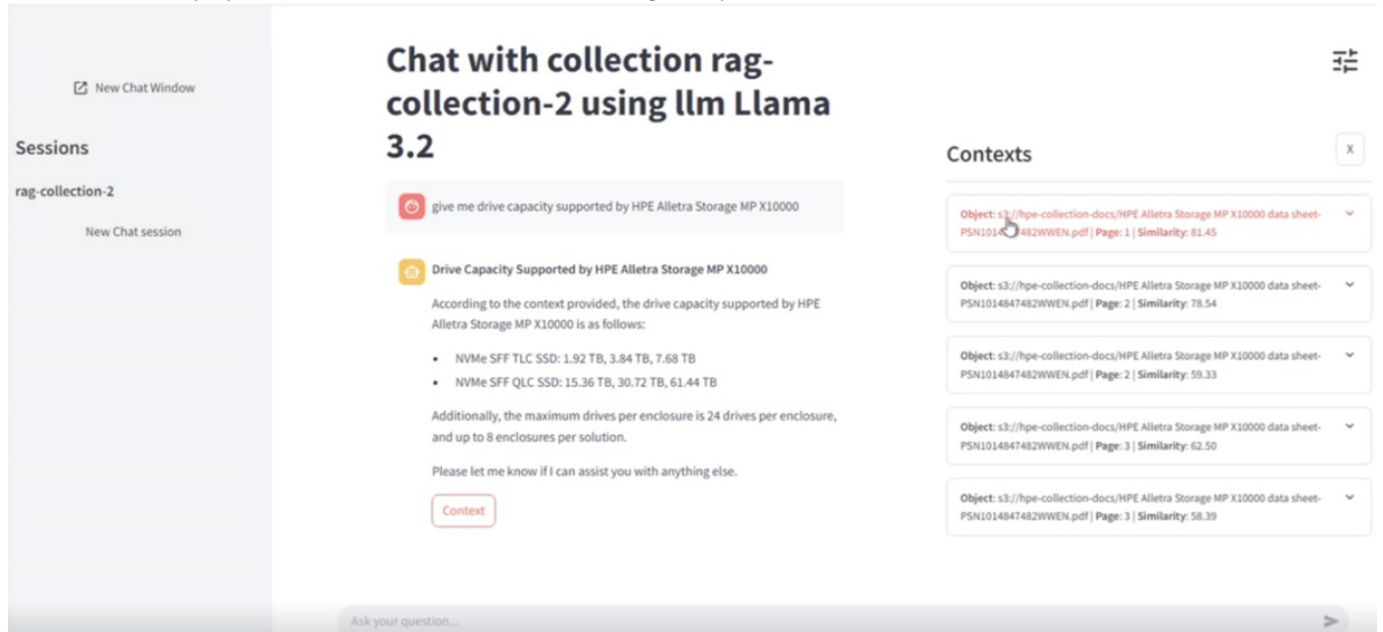
5. Enter the details for the chat:
 - a. Select the LLM (large language model) to use from the dropdown list. The default LLMs are Llama 3.2 and Gemma 3.
 - b. Select the collection to query from the dropdown list.
 - c. Select the number of top-relevant documents to return. You can select a number from 1 to 10.
 - d. Optionally, enter a system prompt to guide the LLM. You can also leave this field blank.
 - e. Click Submit.

A window is displayed, showing the configuring of the chat based on your inputs.



6. Enter a question or directive you want to learn about the data. Click the arrow or press Enter.

The results are displayed on the screen as shown in the following example.



7. Click Context to display the top contexts used to answer the question/prompt. The contexts display on the right-side of the screen. You can select any context to view more information as shown in the following example.

Object: s3://af-rag-demo-bucket/HPE-Alletra-Storage-MP-X10000-QuickSpecs-a50009215enw.pdf | Page: 2 | Similarity: 77.91 ^

QuickSpecs HPE Alletra Storage MP X10000 Overview

Page 2

HPE Alletra Storage MP X10000 (2-Node all NVMe Storage Base)

What is new

- QLC drives for capacity-centric workloads such as data protection – Data intelligence services automatically add custom metadata to objects.
- Optional data protection acceleration (DPA) node(s) to increase performance and efficiency when HPE Alletra Storage MP X10000 provides the storage for high-capacity and high-restore performance data protection workloads. – HPE Alletra Storage MP Disconnected for X10000 is now available

Using Trino CLI (SQL-based access)

About this task

The Trino CLI (Command Line Interface) is an interactive tool for connecting to and querying a Trino cluster. It is mainly used for viewing metadata records. The CLI enables you to connect to a Trino coordinator, run SQL queries interactively or in batch mode, and explore catalogs, schemas, and tables.

Procedure

1. To connect to a Trino coordinator from within a Kubernetes pod, enter the following command:

```
kubect exec -it -n hpe-di <trino-coordinator-pod-name> -- trino
```



NOTE

To identify the `<trino-coordinator-pod-name>`, enter the following command:

```
kubect exec pods -A
```

2. To view all tables, enter the following command:

```
show tables from jdbc.db;
```

3. To view the content of a table, enter the following command:

```
Select * from jdbc.db."collection_name";
```

Example:

```
select * from jdbc.db."659f0676-639c-416c-b359-05f6a4503b6f";
```

The following is an example of commands and output.

```
root@mbs:~# kubectl exec -it nhpe-di r1-trino-coordinator-778f479f9d-h4cth -- trino
Defaulted container "trino-coordinator" out of: trino-coordinator, trino-init (init)
trino>
trino>
trino>
trino> show tables from jdbc.db;
      Table
-----
a09b6284-7e19-4acb-ba9e-500f6435e50
ad51fe13-0ceb-4e38-bf6d-20b0f1f8ce38
collections
models
pipelines
schemas
(6 rows)

Query 20250812_122019_00357_mbtz8, FINISHED, 2 nodes
Splits: 35 total, 35 done (100.00%)
0.05 s (6 rows, 177B) [122 rows/s, 3.53KiB/s]

trino> select * from jdbc.db.collections;
      id          | backend_name          | name      | pipeline
| buckets        | store                | creation_time
-----+-----+-----+-----
87cacf60-e6fa-4f3a-8856-0a8b010168e9 | di_87cacf0e6fa4f3a88650a8b010168e9
| rag-collection-1 | default-rag-pipeline | [di-bucket-1] | milvus-bucket |
2025-08-12 09:37:04.765790
a09b6284-7e19-4acb-baae-500f6435e50 | a09b6284-7e19-4acb-ba9e-
500fe6435e50 | meta-collection-2 | default-metadata-pipeline | [hpe-collection-
docs] | metadata-bucket | 2025-08-12 12:18:48.889807
f68bda98-057a-4b4c-9938-a5855c4e87a1 |
di_f68bd9a805704b4e9938a58855c4e871 | rag-collection-2 | default-rag-pipeline
| [hpe-collection-docs] | milvus-bucket | 2025-08-12 10:31:44.891179
ad51fe13-0ceb-4e38-bf6d-20b0f1f8ce38 | ad51fe13-0ceb-4e38-bf6d-20b0f1f8ce38
| meta-collection-1 | default-metadata-pipeline | [di-bucket-1] | metadata-
bucket | 2025-08-12 10:38:56.645956

Query 20250812_122048_00358_mtbz8, FINISHED, 1 node
Splits: 8 total, 8 done (100.00%)
0.23 s (4 rows, 142KiB) [17 rows/s, 61.5KiB/s]

trino> select * from jdbc.db."a09b6284-7e19-4acb-ba9e-500f6435e50";
      object_key          | bucket          | version_id | size |
last_modified            | content_type    | content_encoding | content_language
-----+-----+-----+-----+-----
HPE Alletra Storage MP X10000 data sheet-PSN1014847482WWEN.pdf | hpe-
collection-docs | NULL        | 467861 | 2025-08-12 12:05:20.000000 | binary/octet-
stream |
(1 row)
```

Subtopics

[Accessing the Milvus database for troubleshooting](#)

[The RAG chatbot is not reachable](#)

[The RAG chatbot does not provide answers](#)

Accessing the Milvus database for troubleshooting

About this task

Use the Milvus database for basic troubleshooting such as:

- Run basic diagnostics
- Inspect metadata
- Identify common operational issues
- Confirm that vector data, indexes, and query nodes are functioning as expected



NOTE

The Milvus database is recommended only for troubleshooting, as vectorDB is managed by the platform.

Access to the database is through the attu UI.

Procedure

1. Create the following `yaml` file (`attu.yaml`), which is required to start the attu services.



NOTE

Creation of the `attu.yaml` file is required for initial access.

```
apiVersion: v1
kind: Service
metadata:
  name: my-attu-svc
labels:
  app: attu
spec:
  type: NodePort
  ports:
    - name: attu
      protocol: TCP
      port: 3000
      targetPort: 3000
  selector:
    app: attu
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-attu
labels:
  app: attu
spec:
  replicas: 1
  selector:
```

```

matchLabels:
  app: attu
template:
  metadata:
    labels:
      app: attu
  spec:
    containers:
      - name: attu
        image: zilliz/attu:v2.5
        imagePullPolicy: IfNotPresent
        ports:
          - name: attu
            containerPort: 3000
            protocol: TCP
        env:
          - name: MILVUS_URL
            value: "r1-milvus.hpe-di.svc:19530"

```

2. To start the attu services in Kubernetes, enter the following command:

```
kubectl apply -f attu.yaml
```

3. (Optional) To verify the service is running, enter the following command:

```
kubectl get svc
```

Command output looks similar to the following:

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	147d
my-attu-svc	NodePort	10.99.237.155	<none>	3000:32001/TCP	9s

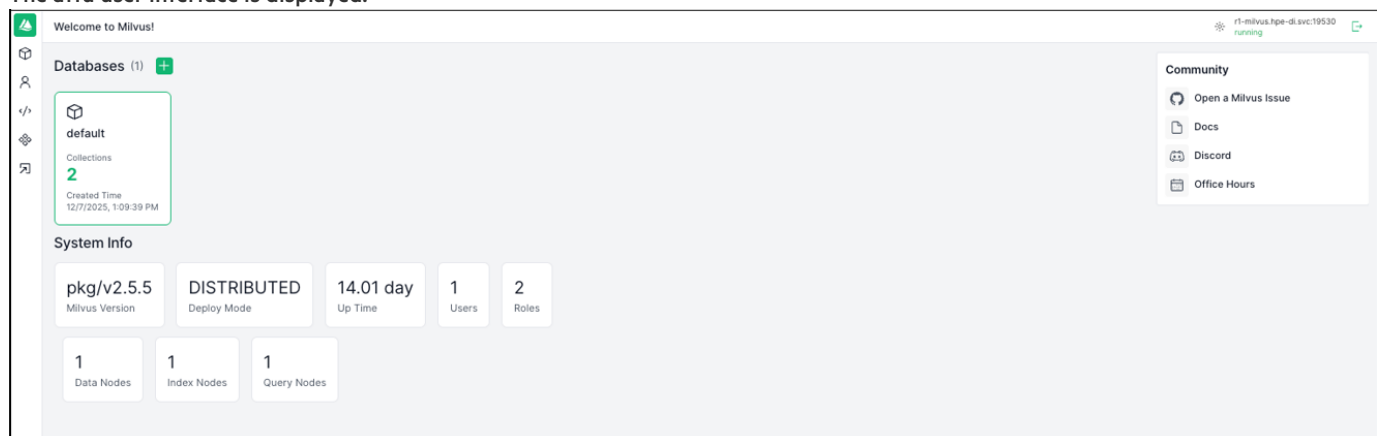
4. Open a browser and enter `http://127.0.0.1:3000`.



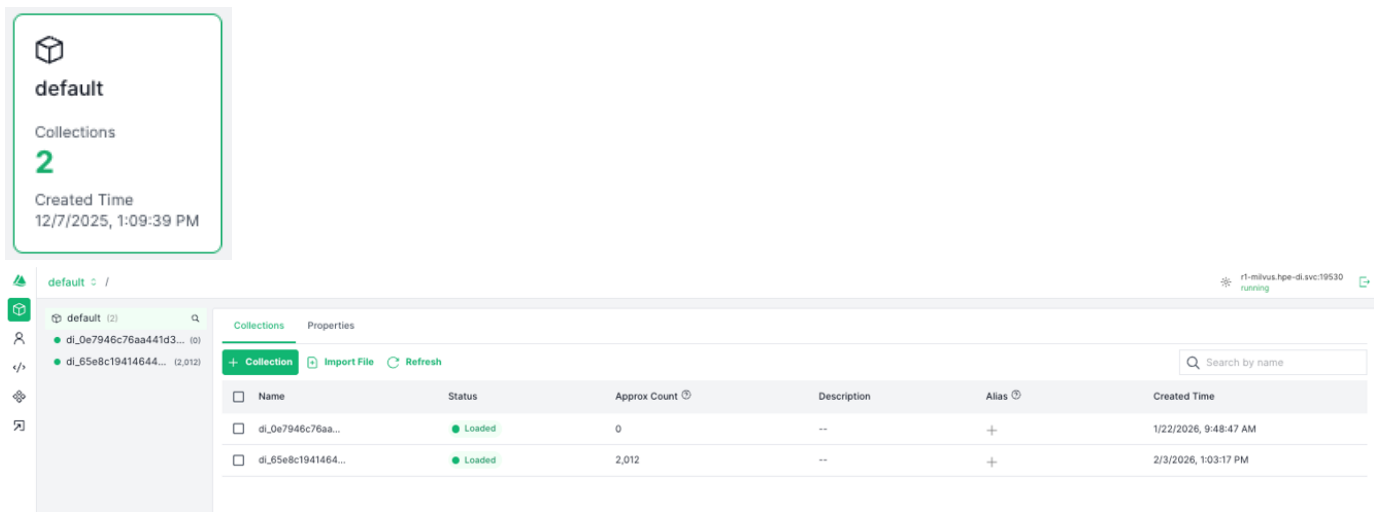
NOTE

- Only `http` is supported with attu.
- If you are accessing the Milvus database from the same server on which it is installed, use ssh: `SSH -L 3000:10.234.33.152:32001 root@10.234.33.152`

The attu user interface is displayed.



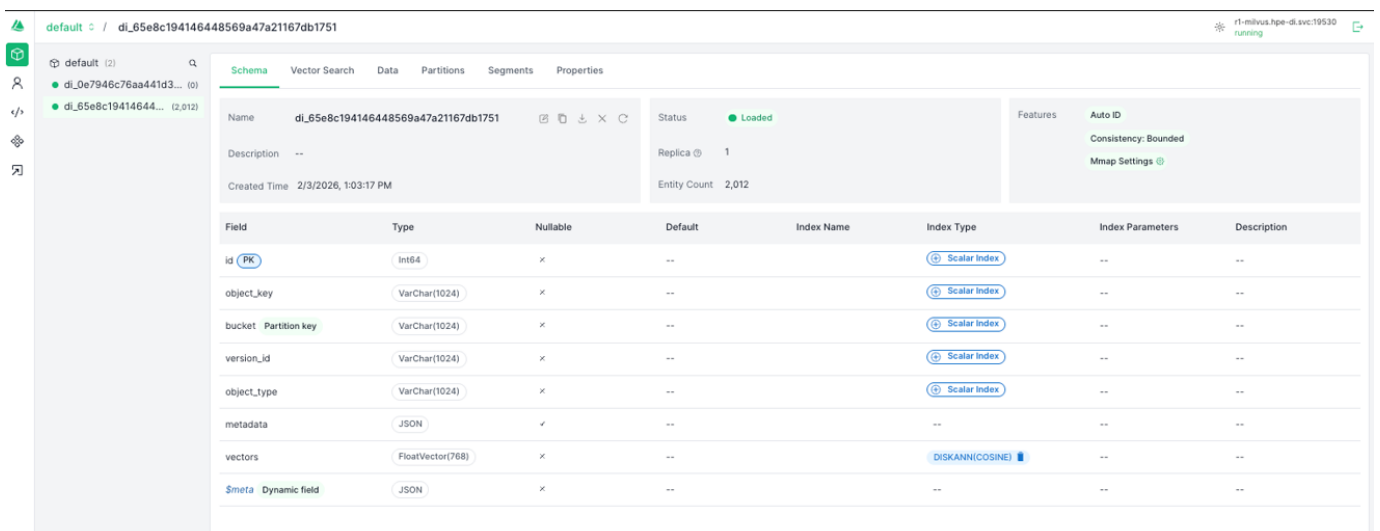
5. Select the default box to display the collections in your database.



The following table describes the columns/fields displayed for the list of collections.

Column/Field name	Description
Name	Name of the collection. It is the same name displayed in the Swagger UI.
Status	Loaded: Data and the index are in RAM and ready for search Unloaded: Data lives only on disk and cannot be searched Load/release: Memory management operations
Approx Count	Estimate of the number of entities (vectors and records)
Description	Description of the collection
Alias	Any alias pointing to this collection. Aliases are logical names used by applications for versioning or swapping. Not used in HPE X10000 Data Intelligence.
Created Time	Time the collection was created

6. Select a collection to display the collection page.



Select from the tabs at the top of the screen to view specific collection details.

Tab	Description
Schema	• Upper section: collection status and main statistics • Bottom section: schema of the collection
Vector search	Search for specific vectors
Data	A list of all the collection's vectors
Partitions	List of the partition in which the collection is divided
Segments	List of the segments in which the collection is divided
Properties	List of the collection's properties

The RAG chatbot is not reachable

Symptom

You cannot access the RAG chatbot. Complete the actions in the following order to determine the issue and resolve.

Solution 1

Action

Verify that service is running. Enter the following command: `kubectl get svc r1-chatbot-app -n hpe-di`

If the service is running, the output of the command should look similar to the following:

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
r1-chatbot-app	NodePort	10.108.114.250	<none>	8501:30501/TCP	58d

Solution 2

Action

Verify that the URL entered in the browser was using http, and not https.

Correct: `http://<Server Name>:30501/`

Incorrect: `https://<Server Name>:30501/`

Solution 3

Action

If you are tunneling into the HPE X10000 Data Intelligence node, be sure to include the `-L` parameter to enforce port forwarding. Use the following ssh command: `ssh -L 8501:<server IP address>:30501 root@<server IP address>`

Use the server IP address, as the chatbot is exposed as NodePort and it is not bound to the 127.0.0.1. port.



The RAG chatbot does not provide answers

Symptom

In response to a question you enter in the RAG chatbot, the chatbot displays the message: `Couldn't find any relevant information in the context.`



Solution 1

Action

Verify that you entered the S3 credentials in the chat configuration screen.

For security reasons, HPE X10000 Data Intelligence binds the chatbot and bucket access together. If you cannot access the bucket (even in read mode), you will not get an answer from the chatbot.

Solution 2

Action

Verify that you selected the correct collection in the chat configuration screen.

If you do not have access to the collection that was selected, you will not get an answer from the chatbot.

Solution 3

Action

In Data Services Cloud Console, verify that you have access to at least one bucket associated with the collection. Navigate to the X10000 dashboard and select Buckets from the menu.

For security reasons, HPE X10000 Data Intelligence binds the chatbot and bucket access together. If you cannot access the bucket (even in read mode), you will not get an answer from the chatbot.

Solution 4

Action

Verify that the collection is populated with vectors embeddings. If the collection is not populated, the system cannot do a search.

1. Access the Milvus database. See [Accessing the Milvus database for troubleshooting](#).
2. Navigate to the applicable collection page.
3. Select the Data tab.

If the collection is populated, the screen will look similar to the following:

id	object_key	bucket	version_id	object_type	metadata	vectors	Dynamic Fields
463718121592832853	Achieving-AI-success-begins-with-a-un...	af-zag-demo-bucket		application/pdf	{"end_char_idx":2157,"page_label":"2"...	[-0.1527179628610611,2.38668537139892...	{}
463718121592832855	Achieving-AI-success-begins-with-a-un...	af-zag-demo-bucket		application/pdf	{"end_char_idx":2556,"page_label":"4"...	[0.8845846692528142,0.332194666133651...	{}
463718121592832857	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1494,"page_label":"1"...	[0.5907394886616846,1.134917259216308...	{}
463718121592832859	Achieving-AI-success-begins-with-a-un...	af-zag-demo-bucket		application/pdf	{"end_char_idx":2157,"page_label":"4"...	[0.6563225984573364,2.96602535248027...	{}
463718121592832861	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1755,"page_label":"2"...	[0.16626816987991333,1.0420496503311...	{}
463718121592832863	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":2280,"page_label":"4"...	[-1.1746151447296143,1.0945222377777...	{}
463718121592832865	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1861,"page_label":"3"...	[0.08663923293352127,1.57608187198638...	{}
463718121592832867	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":2409,"page_label":"4"...	[0.2997922897338867,1.35703659057617...	{}
463718121592832869	DataSheet--HPE-Alletra-Storage-MP-XL...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1296,"page_label":"5"...	[0.529727041721344,0.532150081443786...	{}
463718121592832871	Elastic-Stack-with-HPE-Alletra-Storag...	af-zag-demo-bucket		application/pdf	{"end_char_idx":161,"page_label":"1"...	[0.3907759189605713,1.362217426300048...	{}
463718121592832873	Elastic-Stack-with-HPE-Alletra-Storag...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1715,"page_label":"2"...	[0.6273069977766315,1.91220617294315...	{}
463718121592832874	Elastic-Stack-with-HPE-Alletra-Storag...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1300,"page_label":"3"...	[0.44532525539398193,1.95855259895324...	{}
463718121592832877	Elastic-Stack-with-HPE-Alletra-Storag...	af-zag-demo-bucket		application/pdf	{"end_char_idx":1821,"page_label":"4"...	[0.1752161979675293,1.686980605125427...	{}

If the collection is not populated, the screen will look similar to the following:

id	object_key	bucket	version_id	object_type	metadata	vectors	Dynamic Fields
----	------------	--------	------------	-------------	----------	---------	----------------

No Result

- If the collection is empty but the associated buckets are not, verify that the collection or pipeline configuration is correct.
- If the bucket is empty, you must populate it using a copy command or similar function.

Websites

General websites

Single Point of Connectivity Knowledge (SPOCK) Storage compatibility matrix

<https://www.hpe.com/storage/spock>

Storage white papers and analyst reports

<https://www.hpe.com/storage/whitepapers>

For additional websites, see [Support and other resources](#).

Product websites

Alletra Storage MP X10000 documentation

<https://www.hpe.com/support/x10000docs>

Support and other resources

Subtopics

[Accessing Hewlett Packard Enterprise Support](#)

[HPE product registration](#)

[Accessing updates](#)
[Remote support](#)
[Warranty information](#)
[Regulatory information](#)
[Documentation feedback](#)

Accessing Hewlett Packard Enterprise Support

- For live assistance, go to the Contact Hewlett Packard Enterprise Worldwide website:

<https://www.hpe.com/info/assistance>

- To access documentation and support services, go to the Hewlett Packard Enterprise Support Center website:

<https://www.hpe.com/support/hpesc>

Information to collect

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

HPE product registration

To gain the full benefits of the Hewlett Packard Enterprise Support Center and your purchased support services, add your contracts and products to your account on the HPESC.

- When you add your contracts and products, you receive enhanced personalization, workspace alerts, insights through the dashboards, and easier management of your environment.
- You will also receive recommendations and tailored product knowledge to self-solve any issues, as well as streamlined case creation for faster time to resolution when you must create a case.

To learn how to add your contracts and products, see <https://www.hpe.com/info/add-products-contracts>.

Accessing updates

- Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.
- To download product updates:

Hewlett Packard Enterprise Support Center

<https://www.hpe.com/support/hpesc>

My HPE Software Center

<https://www.hpe.com/software/hpesoftwarecenter>

- To subscribe to eNewsletters and alerts:

<https://www.hpe.com/support/e-updates>

- To view and update your entitlements, and to link your contracts and warranties with your profile, go to the [Hewlett Packard Enterprise Support Center More Information on Access to Support Materials](#) page:

<https://www.hpe.com/support/AccessToSupportMaterials>



IMPORTANT

Access to some updates might require product entitlement when accessed through the [Hewlett Packard Enterprise Support Center](#). You must have an [HPE Account](#) set up with relevant entitlements.

Remote support

Remote support is available with supported devices as part of your warranty or contractual support agreement. It provides intelligent event diagnosis, and automatic, secure submission of hardware event notifications to Hewlett Packard Enterprise, which initiates a fast and accurate resolution based on the service level of your product. Hewlett Packard Enterprise strongly recommends that you register your device for remote support.

If your product includes additional remote support details, use search to locate that information.

HPE Get Connected

<https://www.hpe.com/services/getconnected>

HPE Tech Care Service

<https://www.hpe.com/services/techcare>

HPE Complete Care Service

<https://www.hpe.com/services/completecure>

Warranty information

To view the warranty information for your product, see the [warranty check tool](#).

Regulatory information

To view the regulatory information for your product, view the [Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products](#), available at the Hewlett Packard Enterprise Support Center:

<https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Hewlett Packard Enterprise is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements such as REACH (Regulation EC No 1907/2006 of the European Parliament and the Council). A

chemical information report for this product can be found at:

<https://www.hpe.com/info/reach>

For Hewlett Packard Enterprise product environmental and safety information and compliance data, including RoHS and REACH, see:

<https://www.hpe.com/info/ecodata>

For Hewlett Packard Enterprise environmental information, including company programs, product recycling, and energy efficiency, see:

<https://www.hpe.com/info/environment>

Documentation feedback

Hewlett Packard Enterprise is committed to providing documentation that meets your needs. To help us improve the documentation, click the Feedback button on the page of an opened document on the Hewlett Packard Enterprise Support Center portal

(<https://www.hpe.com/support/hpesc>). Use this feature to send any errors, suggestions, or comments. This process captures all document information.

